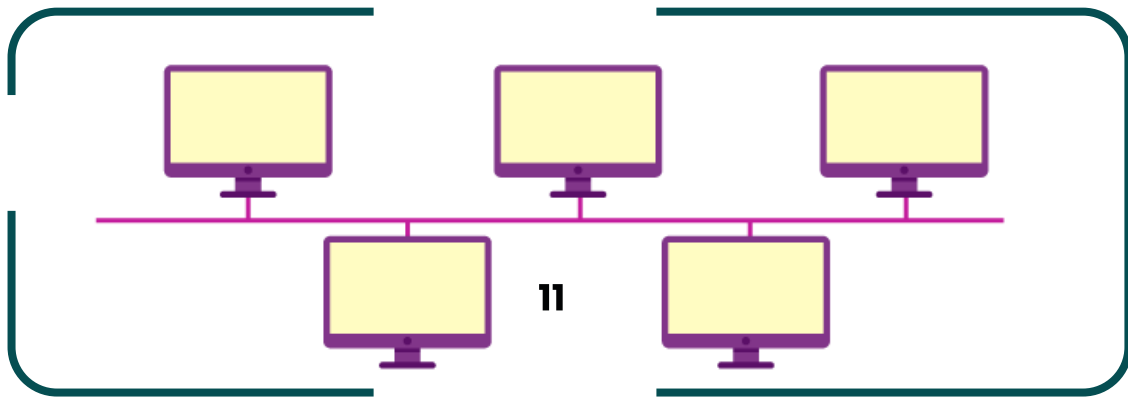


UNIT 2

FUNDAMENTALS OF TOPOLOGIES



Learning Objectives

- Define the term network topology.
- Understand the physical layout of Bus, Ring, and Star topologies.
- Differentiate between various network topologies based on design and functionality of computer.

Networking:

- A network consists of two or more computers that are linked in order to share resources (such as printers and CDs), exchange files or allow electronic communications.
- The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.

Computer Networking

Computer networking refers to the interconnection of multiple devices such as computers, printers, and other hardware to share resources, data, and applications efficiently. These devices are linked through cables, wireless signals, or other means to form networks. Networks can vary in size and complexity, ranging from small home setups to large organizational infrastructures. They allow users to communicate, transfer files, and access shared resources like printers or the internet. Networks can be categorized based on their geographical reach such as Local Area Networks (LANs), Metropolitan Area Networks (MANs), and Wide Area Networks (WANs). The design and layout of these networks, known as topology, are crucial for determining how devices communicate and share information within the network.

Types of Networks:

LAN (Local Area Network)

- Used to connect computers in a small area like a room, school, or office.
- Fast and low-cost.

Example: Computer lab in a school.

MAN (Metropolitan Area Network)

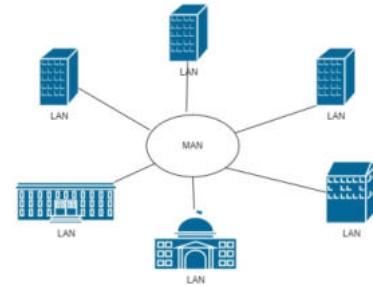
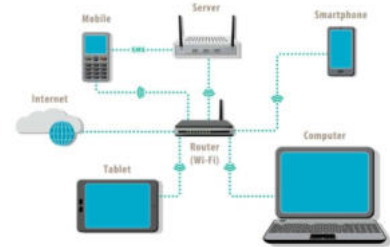
- Covers a city or large campus.
- Bigger than LAN but smaller than WAN.

Example: Connecting all branches of a bank in one city.

WAN (Wide Area Network)

- Connects computers across countries and continents.
- Slower and more expensive than LAN or MAN.

Example: The Internet is the biggest example of WAN.



Fundamentals of Topologies

The physical or logical layout of a network, called topology, defines how devices (nodes) in a network are arranged and how they communicate with one another. Topology is a major consideration when designing networks, as it influences performance, reliability, and scalability.

Types of Topologies

There are several types of network topologies. The most common ones are:

- 1) Bus Topology
- 2) Ring Topology
- 3) Star Topology

1) Bus Topology

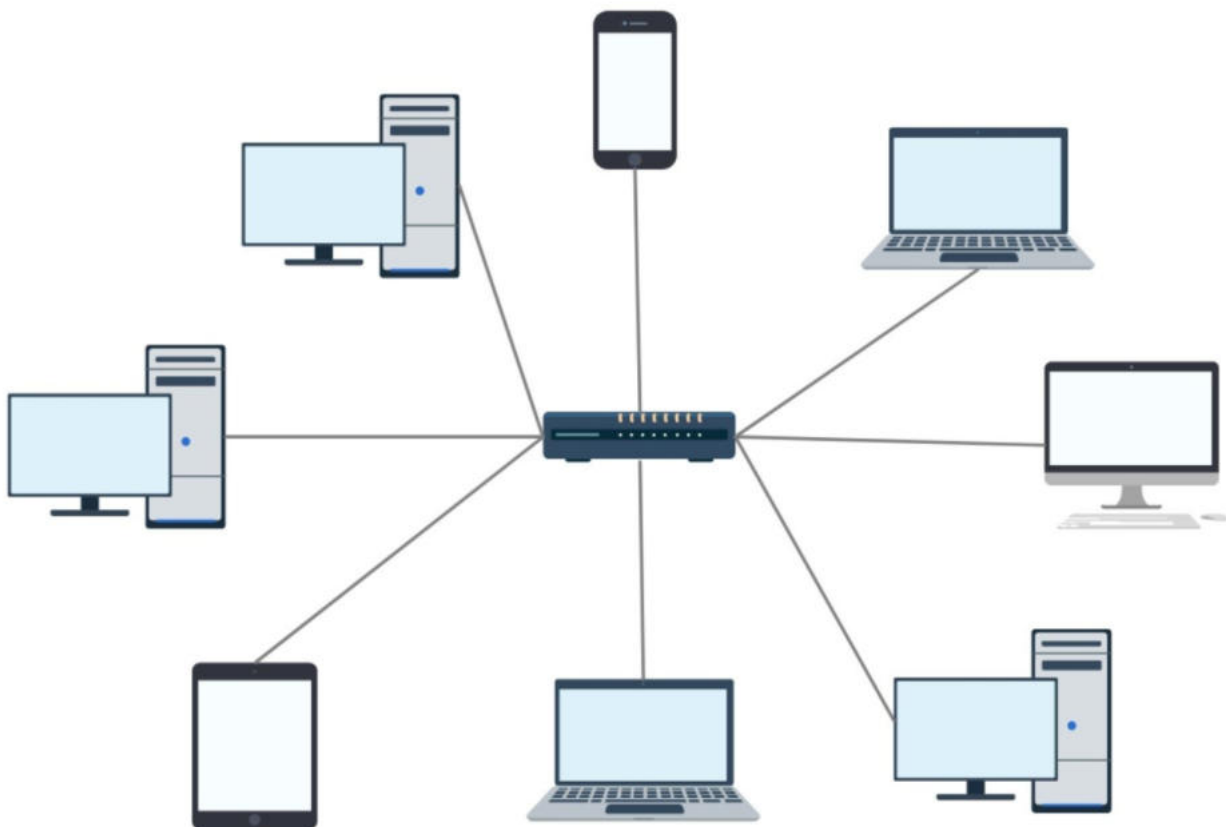
In Bus Topology, all computers and devices are connected through a single cable known as the backbone of the network. Each device communicates with others by sending messages along this cable.

• **Advantages:**

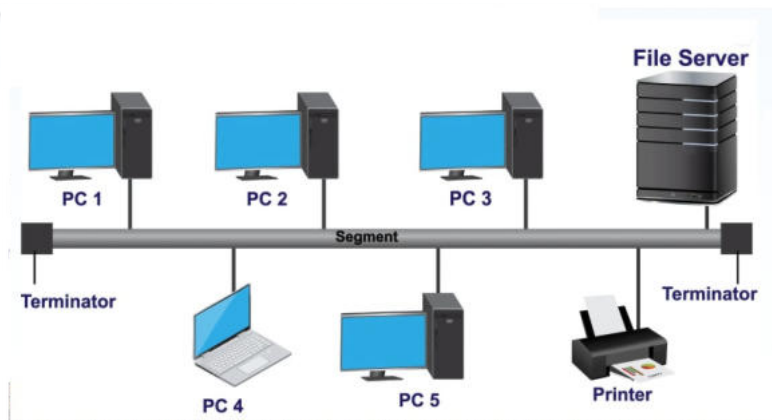
- Simple and cost-effective to implement.
- Easy to expand by adding more devices.

• **Disadvantages:**

- If the backbone cable fails, the entire network goes down.
- Limited scalability due to signal degradation.



Topology



Bus Topology

2) Ring Topology

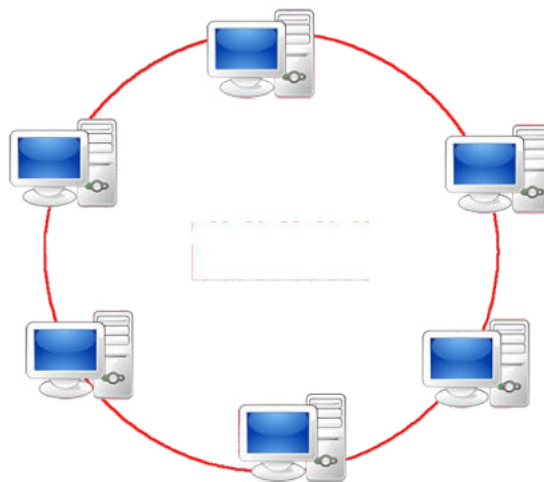
In Ring Topology, devices are connected in a circular layout. Data travels around the loop in one direction, passing through each device until it reaches the intended recipient.

Advantages:

- Easy to troubleshoot because each device has a unique path.
- Can handle high traffic volumes.

Disadvantages:

- Failure of any single device or link can disrupt the entire network.
- Adding or removing devices is more complex than in other topologies.



Ring Topology

3) Star Topology

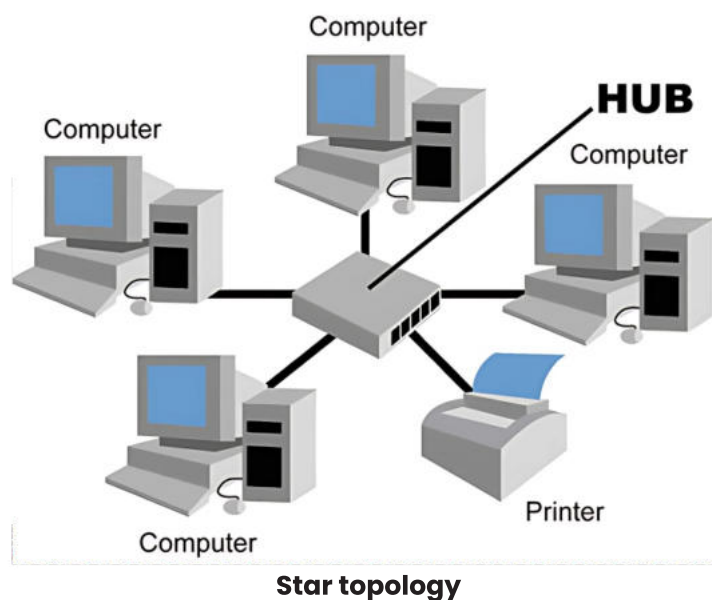
In Star Topology, all devices connect to a central device called a hub or switch. Communication between devices occurs through this central point.

- **Advantages:**

- Easy to set up and expand.
- Failure of one device does not affect the entire network.

- **Disadvantages:**

- If the hub or switch fails, the entire network is disrupted.
- Requires more cable than other topologies.



Understanding the fundamentals of network topologies is essential for designing efficient and reliable computer networks. The choice of topology—whether Bus, Ring, or Star—depends on factors such as cost, scalability, and ease of troubleshooting. Topologies provide the backbone for effective communication and resource sharing in any network. This foundational knowledge prepares students for exploring more advanced topics in networking, such as protocols and network security.

• **Case Study: Choosing the Right Network Topology for ABC Tech Solutions**

Background

ABC Tech Solutions is a growing IT company with 50 employees. Their office consists of three departments: Development, Administration, and Customer Support. Each department requires high-speed internet access, secure data sharing, and reliable communication. The company currently uses individual computers but wants to establish a network to improve collaboration and efficiency.

Requirements:

- Secure and fast data transfer between departments.
- Easy troubleshooting and maintenance.
- Scalability to accommodate future expansion.
- Minimal downtime in case of device failure.

Question:

Based on the given requirements, which network topology would be the most suitable for ABC Tech Solutions? Justify your answer.

EXERCISES

01: Choose the correct answer.

1. What is network topology?

- a) A type of computer hardware
 - b) The physical or logical layout of a network
 - c) A software application
-

2. In which topology do all devices connect to a single backbone cable?

- a) Star
 - b) Ring
 - c) Bus
-

3. What happens if the backbone cable fails in a bus topology?

- a) Only one device stops working
 - b) The entire network goes down
 - c) Data is rerouted automatically
-

4. Which topology connects devices in a circular format?

- a) Bus
 - b) Ring
 - c) Star
-

5. What is the main advantage of star topology?

- a) Uses less cable
 - b) Failure of one device does not affect the entire network
 - c) No central device is required
-

6. In ring topology, how does data travel?

- a) Randomly
- b) In a straight line
- c) In one direction around the loop

7. Which of the following is a disadvantage of ring topology?

- a) Difficult to expand
 - b) Network fails if one node fails
 - c) Both a and b
-

8. In which topology a hub or switch is used?

- a) Ring
 - b) Bus
 - c) Star
-

9. What type of network is best for handling high traffic?

- a) Bus
 - b) Ring
 - c) Star
-

10. Which topology is the simplest and most cost-effective to implement?

- a) Ring
 - b) Bus
 - c) Star
-

02: Fill in the blanks.

1. _____ is the arrangement of devices in a network.
2. A bus topology uses a single _____ as the main connection.
3. In ring topology, data moves in _____ direction.
4. A _____ or switch is required in a star topology.
5. If the hub in a star topology fails, the _____ network stops working.
6. The physical layout of a network affects its _____ and scalability.
7. The main disadvantage of bus topology is that if the main cable fails, the _____ network stops functioning.
8. In a _____ topology, each device is connected in a closed loop.

03: Write True or False for each statement.

	TRUE	FALSE
1. Network topology refers to the design and layout of a network.	☐	☐
2. Star topology is the least reliable network topology.	☐	☐
3. In bus topology, failure of one device does not affect the network.	☐	☐
4. A ring topology is best for handling high traffic loads.	☐	☐
5. In a star topology, all devices are connected directly to each other	☐	☐
6. The backbone cable in a bus topology carries all data.	☐	☐
7. A hub is an essential part of ring topology.	☐	☐
8. Adding or removing devices in a ring topology is simple.	☐	☐

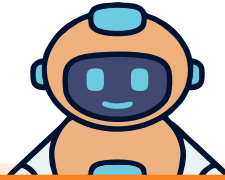
04: Answer the following questions briefly:

1. Define network topology and its importance in networking.
2. How does a bus topology function?
3. What is the role of a hub or switch in a star topology?
4. Write one advantage of star topology.
5. Why is bus topology considered cost-effective?
6. How does topology affect network scalability?

05: Provide detailed answers of the following questions:

1. Compare and contrast bus, ring, and star topologies with examples.
2. Which network topology would be best for a large office? Justify your answer.
3. Describe the advantages of bus, star and ring topology.
4. How does the choice of network topology impact network security and efficiency?

Lab Activities:



1. Identifying Network Topologies (Hands-on Activity)

Objective: Understand different network topologies by arranging physical or virtual connections.

Instructions:

- Use string or network cables to physically demonstrate Bus, Ring, and Star topologies on a table.
- Assign students different roles (computer, hub, cable) and simulate data transmission by passing messages.
- Compare the strengths and weaknesses of each topology.

2. Simulating Network Topologies using Packet Tracer

Objective: Learn how to design and test network topologies using simulation software.

Instructions:

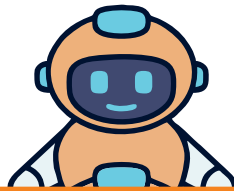
- Open Cisco Packet Tracer or another network simulation tool.
- Create Bus, Ring, and Star topology diagrams using virtual network devices.
- Test data transmission by sending packets between computers and observing how the network behaves.

3. Troubleshooting a Faulty Network Topology

Objective: Identify and resolve network issues based on given scenarios.

Instructions:

- Set up a small network in a lab using actual networking hardware (switches, routers, and cables) or simulate it using Packet Tracer.
- Introduce a fault, such as disconnecting a cable in Bus topology or shutting down a hub in Star topology.



- Have students diagnose and suggest solutions for the network failure.

4. Compare Real-world Network Setups

Objective: Analyze and compare the network topology of different environments.

Instructions:

- Visit the school's computer lab and sketch the existing network topology.
- Research or interview IT staff about the advantages and limitations of the current setup.
- Compare the findings with a home network and a corporate office setup.

5. Designing a Network for a Small Business

Objective: Apply knowledge of topologies to a real-world scenario.

Instructions:

- Give students a case study of a small business needing a network.
- Have them design a network layout using appropriate topology and justify their choice.
- Present their designs to the class and discuss the advantages and disadvantages.